

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER ARCHITECTURE COURSE CODE: CSA12

COURSE OUTCOME COVERED

CO1: Analyze the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking (BL4)
Assessment Tool Weightage: 40%

QUESTIONS:5

Total Marks:25

ANNEXURE A APPLICATION BASED QUESTIONS

Code of Conduct:

*I R.MOHAN RAM / 12521117 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

ANNEXURE A

1. A smart city surveillance system processes continuous video feeds from multiple cameras, where large volumes of data are transferred between I/O devices, memory, and CPU. During peak hours, the system experiences lag and delayed response in threat detection. Analyze how the interaction between CPU, memory, and I/O contributes to this issue, and explain how different bus structures (single-bus vs multi-bus) and improvements in instruction execution cycle can enhance system performance.
2. A real-time stock trading application running on a processor executes millions of instructions per second, but users report delays in transaction processing during high load. The system has a high CPI due to frequent memory access and branch instructions. Examine how the fetch-decode-execute cycle impacts execution time in this scenario, and propose methods to reduce CPI and improve overall CPU performance using suitable architectural enhancements.
3. An embedded system based on an 8085 microprocessor is used in an industrial temperature control unit, where sensors provide input and actuators respond accordingly. However, incorrect temperature readings are occasionally processed due to improper use of addressing modes in the program. Discuss how different addressing modes influence data access, identify possible causes of such errors, and suggest how the instruction set of 8085 can be effectively used to ensure accurate and reliable operation.

4. A computing system uses an 8086 microprocessor with segmented memory for executing multiple programs simultaneously, but it encounters issues like incorrect data access and stack overflow errors. Evaluate how segmentation (code, data, and stack segments) works in 8086, analyze how improper segment handling can lead to such problems, and explain how instruction execution and addressing modes must be managed to ensure correct program execution.
5. A data communication system represents packet information in hexadecimal format for ease of debugging and transmission, but a software engineer mistakenly interprets values, leading to incorrect data processing. Analyze the importance of number system conversions between hexadecimal and binary in such systems, explain how errors in conversion can affect instruction execution, and discuss how efficient CPU design and benchmarking can help detect and minimize such issues in real-time systems.

MARKS ALLOCATION

Criteria	Understanding of Problem Context (20%)	Application of Concepts (20%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (20%)
Q1					
Q2					
Q3					
Q4					
Q5					

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

① Smart City Surveillance CPU, Memory, I/O & Bus.

⇒ In this system, I/O devices (Cameras) send large Video data to memory, and CPU Processes it.

⇒ During peak time, too much data causes bottleneck between CPU, memory, and I/O → leads to lag and delay.

~~Problem~~ Problem Reason.

⇒ Single bus → all devices share same path → traffic Jam

⇒ CPU waits for memory → delay increases.

Solution:

⇒ Multi-bus system - separate buses → faster data transfer.

⇒ cache memory → reduces memory access time.

⇒ pipe line execution → faster instruction processing.

② Stock Trading App - CPI 4 Execution 8 cycle.

- ⇒ CPU works in Fetch → Decode → Execute cycle
- ⇒ High CPI means more clock cycle per instruction, so execution becomes slow.

Problem :

- ⇒ frequent memory access.
- ⇒ Many branch instructions.

Solution.

- ⇒ Cache memory → reduce memory delay.
- ⇒ Branch prediction → avoid waiting.
- ⇒ pipelining → parallel execution.
- ⇒ RISC architecture → fewer cycle per instruction.

③ 8085 Addressing modes Error.

⇒ Addressing modes decide how data is accessed (direct, indirect, immediate).

Problem

⇒ Wrong addressing mode → incorrect data read.

⇒ Example: using direct instead of indirect

Solution

⇒ Use correct addressing mode.

⇒ ~~tt~~ Verify memory locations.

⇒ Use proper 8085 instructions.

④ 8086 Segmentation problem

⇒ 8086 divides memory into:

→ Code Segment.

→ Data Segment.

→ Stack Segment.

Problem:

⇒ wrong segment usage → incorrect data.

⇒ Stack overflow → too much push without pop.

Solution:

⇒ proper segment register usage (CS; DS, SS).

⇒ Manage stack properly.

⇒ correct addressing modes.

⑤ Hexadecimal & Binary Conversion

→ Computers work in binary, but humans use hexadecimal for simplicity.

Problem :

- Wrong conversion → wrong data processing.
- leads to incorrect instruction execution.

Solution

- Accurate hex ↔ binary conversion.
- use validation tools.
- Benchmark CPU performance to detect errors.